

Linux Fundamentals: User Management Hands-on Assignments

Assignment 1: Basic User Account Provisioning

Production scenario: A new junior developer has joined the application team and needs a standard Linux login account with a home directory and Bash shell.

- Create a user named devuser01 with a home directory and default shell set to /bin/bash.
- Set a temporary password for the user.

```
root@localhost:~/vi_assignment03# cd ..
root@localhost:~# sudo useradd -m -s /bin/bash devuser01
root@localhost:~# ls -ld /home/devuser01
drwx-----. 2 devuser01 devuser01 93 Aug 23 16:16 /home/devuser01
root@localhost:~# sudo passwd devuser01
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
root@localhost:~#
```

- Verify the user account using id, groups, and the relevant entry in /etc/passwd.

```
passwd: password updated successfully
root@localhost:~# id devuser01
uid=6667(devuser01) gid=6667(devuser01) groups=6667(devuser01)
root@localhost:~# groups devuser01
devuser01 : devuser01
root@localhost:~# grep '^devuser01:' /etc/passwd
devuser01:x:6667:6667::/home/devuser01:/bin/bash
root@localhost:~# _
```

- Switch to the new user and create a file named welcome.txt inside the home directory.
- Record all commands used and screenshots or terminal output as evidence.

Expected outcome: The student can safely create a regular Linux user, assign a password, confirm UID/GID details, and validate login usability without directly editing system account files.

```
devuser01@localhost:~$ su - devuser01
root@localhost:~# su - devuser01
devuser01@localhost:~$ whoami
devuser01
devuser01@localhost:~$ pwd
/home/devuser01
devuser01@localhost:~$ touch welcome.txt
devuser01@localhost:~$ ls -l welcome.txt
-rw-r--r--. 1 devuser01 devuser01 0 Aug 23 16:43 welcome.txt
devuser01@localhost:~$ echo "Welcome to linux, devuser01" > welcome.txt
devuser01@localhost:~$ cat welcome.txt
Welcome to linux, devuser01
devuser01@localhost:~$ exit
logout
root@localhost:~# _
```

Assignment 2: User Modification with usermod

Production scenario: An existing employee has moved from a general support role to the application operations team, and the Linux administrator must update the account without deleting or recreating it.

- Create a user named support01 with a home directory and default Bash shell.

```
devuser01@localhost:~$ exit
logout
root@localhost:~# sudo useradd -m -s /bin/bash support01
root@localhost:~# id support01
uid=6668(support01) gid=6668(support01) groups=6668(support01)
root@localhost:~# grep '^support01:' /etc/passwd
support01:x:6668:6668::/home/support01:/bin/bash
root@localhost:~# sudo passwd support01
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
root@localhost:~#
```

- Modify the user's comment field to include the full name and department, such as Support User, Application Operations.

```
useradd: user support01 already exists
root@localhost:~# usermod -c "Support User, Application Operations" support01
root@localhost:~# grep '^support01:' /etc/passwd
support01:x:6668:6668:Support User, Application Operations:/home/support01:/bin/bash
root@localhost:~#
```

- Change the user's login shell to /bin/sh, then verify the change in /etc/passwd.

```
root@localhost:~# sudo usermod -s /bin/sh support01
root@localhost:~# grep '^support01:' /etc/passwd
support01:x:6668:6668:Support User, Application Operations:/home/support01:/bin/sh
```

- Create a new home directory path named /home/appsupport01 and move the existing home directory contents to the new location using the correct user modification option.

```
root@localhost:~# sudo usermod -d /home/appsupport01 -m support01
root@localhost:~# grep '^support01:' /etc/passwd
support01:x:6668:6668:Support User, Application Operations:/home/appsupport01:/bin/sh
root@localhost:~#
```

- Create a group named appops and add support01 as a supplementary group member without removing any existing group memberships.

```
root@localhost:~# sudo groupadd appops
root@localhost:~# getent group appops
appops:x:6669:
root@localhost:~# sudo usermod -aG appops support01
root@localhost:~# id support01
uid=6668(support01) gid=6668(support01) groups=6668(support01),6669(appops)
```

- Lock the user account, verify that the account is locked, then unlock it and verify normal access again.

```
root@localhost:~# sudo usermod -L support01
```

```
root@localhost:~# sudo passwd -S support01
support01 L 2026-08-23 0 99999 7 -1
root@localhost:~# sudo usermod -U support01
root@localhost:~# sudo passwd -S support01
support01 P 2026-08-23 0 99999 7 -1
root@localhost:~#
```

- Document each before-and-after value, including username, UID, home directory, shell, group membership, and lock status.

Expected outcome: The student understands how to safely modify existing Linux user accounts using usermod, including updating metadata, shell, home directory, supplementary groups, and lock status while preserving production account continuity.

```
root@localhost:~# id support01
uid=6668(support01) gid=6668(support01) groups=6668(support01),6669(appops)
root@localhost:~# grep '^support01:' /etc/passwd
support01:x:6668:6668:Support User, Application Operations:/home/appsupport01:/bin/sh
root@localhost:~# groups support01
support01 : support01 appops
root@localhost:~# passwd -S support01
support01 P 2026-08-23 0 99999 7 -1
root@localhost:~# su - support01
Last login: Sun Aug 23 17:34:19 IST 2026 on tty1
support01@localhost:~$ whoami
support01
support01@localhost:~$ pwd
/home/appsupport01
support01@localhost:~$ id
uid=6668(support01) gid=6668(support01) groups=6668(support01),6669(appops) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
support01@localhost:~$
```

Assignment 3: User Password and Account Aging Policy

Production scenario: A security team has asked the Linux administrator to enforce password aging and account expiry controls for temporary and permanent users on a production server.

- Create a user named contractor01 with a home directory and Bash shell.
- View the current password aging settings for contractor01 using the appropriate command.
- Configure password aging so the user must keep a password for at least 7 days, must change it after 90 days, and receives a warning 14 days before expiry.
- Set the account to expire on a specific future date to simulate a contract end date.

```

root@localhost:~# useradd -m -s /bin/bash contractor01
root@localhost:~# id contractor01
uid=6669(contractor01) gid=6670(contractor01) groups=6670(contractor01)
root@localhost:~# getent passwd contractor01
contractor01:x:6669:6670::/home/contractor01:/bin/bash
root@localhost:~# passwd contractor01
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
root@localhost:~# chage -l contractor01
Last password change                : Aug 24, 2026
Password expires                    : never
Password inactive                   : never
Account expires                    : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
root@localhost:~# chage -m 7 contractor01
root@localhost:~# chage -M 90 contractor01
root@localhost:~# chage -W 14 contractor01
root@localhost:~# chage -E 2026-12-31 contractor01
root@localhost:~# chage -l contractor01
Last password change                : Aug 24, 2026
Password expires                    : Nov 22, 2026
Password inactive                   : never
Account expires                    : Dec 30, 2026
Minimum number of days between password change : 7
Maximum number of days between password change : 90
Number of days of warning before password expires : 14

```

- Configure an inactive period so the account is disabled if the password remains expired for 30 days.

```

root@localhost:~# chage -I 30 contractor01
root@localhost:~# chage -l contractor01
Last password change                : Aug 24, 2026
Password expires                    : Nov 22, 2026
Password inactive                   : Dec 22, 2026
Account expires                    : Dec 30, 2026
Minimum number of days between password change : 7
Maximum number of days between password change : 90
Number of days of warning before password expires : 14
root@localhost:~#

```

- Force the user to change the password at the next login and verify the effect.

```

root@localhost:~# chage -d 0 contractor01
root@localhost:~# chage -l contractor01
Last password change                : password must be changed
Password expires                    : password must be changed
Password inactive                   : password must be changed
Account expires                    : Dec 30, 2026
Minimum number of days between password change : 7
Maximum number of days between password change : 90
Number of days of warning before password expires : 14

```

- Compare the user-specific aging settings with the system-wide defaults in /etc/login.defs, and explain whether the defaults apply to existing or only newly created users.

```

root@localhost:~# grep PASS /etc/login.defs
# PASS_MAX_DAYS Maximum number of days a password may be used.
# PASS_MIN_DAYS Minimum number of days allowed between password changes.
# PASS_MIN_LEN Minimum acceptable password length.
# PASS_WARN_AGE Number of days warning given before a password expires.
PASS_MAX_DAYS 99999
PASS_MIN_DAYS 0
PASS_MIN_LEN 8
PASS_WARN_AGE 7
PASS_CHANGE_TRIES 5
PASS_ALWAYS_WARN yes
#PASS_MAX_LEN 8
root@localhost:~#

```

- Prepare production evidence showing the before-and-after aging policy, commands used, and validation output.

```

root@localhost:~# chage -m 7 -M 90 -W 14 contractor01
root@localhost:~# chage -E 2026-12-31 contractor01

root@localhost:~# chage -I 30 contractor01
root@localhost:~# chage -d 0 contractor01
root@localhost:~# chage -l contractor01
Last password change           : password must be changed
Password expires               : password must be changed
Password inactive              : password must be changed
Account expires                : Dec 30, 2026
Minimum number of days between password change : 7
Maximum number of days between password change : 90
Number of days of warning before password expires : 14
root@localhost:~#

```

Expected outcome: The student understands password maximum age, minimum age, warning period, inactivity lock, account expiry date, forced password change, and how these controls support production security and compliance requirements.